tips, tricks, ethics

adam okulicz-kozaryn

adam.okulicz.kozaryn@gmail.com

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outline

tips and tricks

ethics

research design: important from now on

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join data

- the real value comes from joining data!
- again, a map about any single var has already been made
  - but 2 or more vars from varied sources: rare
- ton of datasets and vars out there!
  - use your creativity and imagination
  - and you'll easily come up with something that no one did

have a big screen

- a big screen is key for gis work
- around $250
  - will get 30-100% more productive
  - return on investment in couple weeks, at min wag $15/hr
  - not just size but resolution matters!
  - >= 4K, say UHD 3840x2160
- and get a nice mouse too

take your work to the next level

- do not keep these maps in a drawer, share with the world
  - show to your coworkers!
  - send to local newspaper; post on your website and social media
- go scholarly! really the only way to take it to the top
  - present at conferences
  - read the literature on google scholar
  - and email the authors and ask them to colaborate
  - (but have to be advanced and already done research yourself)

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actually important

- not just some second thought sidenote
- fine, do consulting, make $, don't be broke
  - but add: potential/perceived conflict of interest/funding
  - and do at least some work that's independent

integrity/honesty

- be explicit about problems in your data
  - eg non-joins, missing data, miscodings
- be explicit about holes in your story:
  - eg don't hide maps bc they contradict your story
  - discuss: how, why; ask audience to comment/criticize
- dont force data to tell your story:
  - even be devil's advocate: try to find holes yourself
  - listen carefully; let data tell you her story!

ethics

- everybody wants to sell something
- including academics, thinkers, and students!
  - we try to sell some idea or point of view
- nobody's 100% objective
- try to present alternative/opposite points of view
- present the whole picture
- do force yourself to be objective, because humans aren't
- see eye-opening `righteousmind.com`

bad examples

- cherry picking of vars or samples or timeframes, etc
  - eg using only vars/operationalizations that fit your story
  - eg using year in which you find what you wanted to find
  - classification: playing with bins to fit your story

AI

- Yes do use it!
- But disclose, and do be specific how you used it!
  - ideally give engine and prompt, eg
  - gemini “make thematic map of population”
- do not overuse it! eg
  - you feed syllabus and ps into it and ask to produce ps on such and such topic
  - it may be good ps, and ai gets an A, but you F or C

AI

- again need to cite ai
  - (and dont overuse it–i take points off)
- its good to use ai, its a skill, and great tool, but need to cite it and it must be clear whats yours and whats ai's
  - dont say 'used claude'
    give me prompts and output, full log
  - one way to move forward is to have 2 notebooks: one fully ai and the other fully yours
- more is less: in general i'd ditch some maps, you have pleny, just pick the best ones
- you have enough here, cut back, think it through, interpret, tell a story–whats the overall main bottomline

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quality

- GIGO: Garbage In, Garbage Out
- 'Cos it's in the computer, don't mean it's right
  - double, triple check
  - ask yourself if it makes sense...
  - (Camden richer than Cherry Hill?)
- and who produced the data? Trust US or Indian govt? Bias in what direction?

construct validity; seven sins

- are you measuring what you say you are measuring?
- say want IQ, but you only have education data
- http://www.socialresearchmethods.net/kb/constval.php
- http://2.bp.blogspot.com/_R3SXJVojagU/SwLzZJL1E2I/AAAAAAAAAIE/7GbMzcZPDDk/s1600/sevendeadlysins.bmp
  - https://creativeclass.com/whos-your-city/maps/#Personality_Maps
  - https://scholar.google.com/citations?user=TZ9YLMoAAAAJ&hl=en&oi=ao
- https://creativeclass.com/whos-your-city/maps/#The_Singles_Map
  - how measure singles? polygamy, cheating, etc
- much downwards bias too eg: crime, vote (shy Trump)

triangulate

- use several datasources and or several variables to measure the same thing (triangulation)
- triangulation=use different measures for the same concept
- eg education:
- years of schooling
- highest degree obtained
- avg SAT score
- avg ranking of schools in the area
- etc etc

external validity

- are your data representative ?
- how big is the sample ?
- eg I was geocoding WVS at province level only to find out it was unrepresentative